

**Ilias Test:
Observation →**

Question 1



Which of the following are considered "best practices" or guidelines for the design of a usability questionnaire? (Select all that apply)

- Right: Avoid jargon and language that the interviewee may not understand.
- Wrong: Use very long questionnaires to ensure every possible detail is covered.
- Right: Provide clear instructions on how to complete the questionnaire.
- Right: Split compound sentences into two separate questions.
- Right: Give the option for "free" comments at the end.

Question 2



What is a central advantage of ESM compared with a single retrospective questionnaire?

- Right: It captures experience closer to the situation.
- Wrong: It eliminates participant burden completely.
- Wrong: It requires no questions.
- Wrong: It only works in laboratory studies.
- Wrong: It prevents event-based sampling.

Question 3



Match the ethnographic element to its meaning:

- Participant observation -> Researcher observes while engaging with context
- Interviews -> Method for eliciting participant perspectives
- Immersion in culture -> Experiencing participant and context from within
- Comments, incidents, artifacts -> Materials collected for analysis

Question 4



What are the characteristics or goals of "Experience Studies" within Ubicomp field studies? (Select all that apply)

- Right: Understanding how technology use changes people's behavior over a long period.
- Wrong: Verifying if a new technology functions technically in the real world.
- Right: Using the Experience Sampling Method (ESM) to collect data throughout the day.
- Wrong: Conducting a short, 15-minute controlled lab session.

Question 5



In expert inspections without users, which problems are highlighted as major risks?

- Right: Important problems may be missed.
- Right: Many trivial problems are often identified.
- Right: Expert biases can distort problem prioritization.
- Wrong: Detailed user journey documentation is required.
- Wrong: Findings cannot be replicated by other experts.

Question 6



Match the evaluation method to the primary type of human memory it accesses based on the lecture's memory model:

- Think Aloud Protocol -> Working Memory
- Interviews asking for recall of a past situation -> Episodic Memory
- Questionnaires asking for general facts -> Semantic Memory
- Observation of practiced task execution -> Procedural Memory

Question 7



Within an iterative design process, what is the specific relationship between a summative evaluation of version 'n' and a formative evaluation of version 'n+1'?

- Wrong: They are independent processes that should not overlap.
- Right: A summative evaluation of one design often serves as a formative evaluation for the next development step.
- Wrong: Formative evaluation must always precede summative evaluation in every single iteration.
- Wrong: Summative evaluation is only used for final products and cannot inform future designs.

Question 8



Match the criterion for creating a measure of mental workload to its corresponding methodological concern:

- Reliability -> Same estimate for same task and operator
- Diagnosticity -> Indicates cause of workload variation
- Selectivity -> Avoids influence of unrelated factors
- Sensitivity -> Detects changes in task demand

Question 9



What is the primary methodological purpose of the second step in the NASA Task Load Index (NASA-TLX), which involves 15 pairwise comparisons?

- Wrong: To increase the number of data points for better statistical significance.
- Right: To weight the workload dimensions according to which factors the participant considers most important for the specific task.
- Wrong: To verify that the participant provided consistent answers in the first step.
- Wrong: To transform subjective Likert-scale results into objective physical measurements.

Question 10



Match the sampling strategy to its correct definition.

- Random Sampling -> Everyone in a list has an equal probability of being selected.
- Stratified Sampling -> Selection ensures participants reflect the distribution (e.g., gender) of the target population.
- Systematic Sampling -> Participants are chosen based on a predefined criterion, such as every 10th person.
- Sample of Convenience -> Participants are selected because they are easily available.

Question 11



Which of the following outcomes are mentioned in the lecture as indicators associated with increased workload in secondary-task performance? (Select all that apply)

- Right: Increased variability in a rhythmic tapping task.
- Right: A decline in the degree of randomness in a random number generation task.
- Right: Increased reaction time to a probe stimulus.
- Right: Reduced accuracy in the secondary task.

Question 12



Match the Ubicomp field study type to its corresponding result or goal:

- Studies of current behavior -> Better understanding of current technology use and future implications
- Proof-of-concept studies -> Validation of whether a new technology functions in the real world
- Experience studies -> Understanding behavioral change over a long period (weeks/months)

Question 13



If a researcher is concerned that their laboratory conditions do not sufficiently reflect the "real-world" context of the users, which specific type of validity is being questioned?

- Wrong: Internal Validity
- Wrong: External Validity
- Right: Ecological Validity
- Wrong: Construct Validity

Question 14



When designing a study, which of the following are considered "extraneous" or "confounding" variables that can distort results if not controlled through randomization, matching, standardization, or procedural controls? (Select all that apply)

- Right: The participant's prior knowledge about the system being tested.
- Right: Environmental factors like room temperature or noise levels during an interview.
- Wrong: The independent variable manipulated by the researcher.
- Right: The attitude or bias of the interviewer toward the system.